

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JNANA SANGAMA”, BELAGAVI - 590018



DEEP LEARNING MINI PROJECT REPORT(CS722T2C)

on

“Speech Emotion Recognition using Deep Learning”

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In partial fulfillment of the requirements for the VII semester

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE & ENGINEERING

Under the Guidance of

Ms. Shiji Abraham

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at



SAHYADRI

College of Engineering & Management

An Autonomous Institution

MANGALURU

2026- 27



EVALUATION RUBRICS

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Sl. No.	Evaluation Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)	Marks
1	Problem Definition & Objectives	Problem is clearly defined, relevant, and objectives are specific and measurable	Problem and objectives are clear with minor gaps	Problem is partially defined; objectives need improvement	Problem/objectives are unclear	
2	Technical Approach & Implementation	Appropriate methodology, algorithms/tools used effectively; implementation is complete	Appropriate technical approach with minor issues	Basic implementation with limited technical depth	Incomplete or inappropriate implementation	
3	Innovation / Application / Problem Solving	Demonstrates strong originality, creativity, and effective solution to the identified problem	Shows some innovation and good problem-solving	Limited innovation; mostly conventional approach	No significant innovation/problem-solving	
4	Results, Testing & Analysis	Results are clearly presented, thoroughly tested, and critically analyzed using suitable measures	Results presented with adequate testing and analysis	Basic results and limited testing/analysis	Results are incomplete or unsupported	
5	Documentation, Presentation & Viva	Well-structured report, clear presentation, excellent communication and confident viva	Good documentation, presentation and satisfactory viva	Adequate documentation/presentation with limited clarity	Poor documentation, presentation or viva	
	Total					

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Department of Computer Science & Engineering

DECLARATION

We hereby declare that the entire work embodied in this Mini Project Report titled “**Speech Emotion Recognition using Deep Learning**” has been carried out by us at Sahyadri College of Engineering & Management, Mangaluru under the supervision of **Ms. Shiji Abraham**, in partial fulfillment of the requirements for the VII semester of **Bachelor of Engineering in Computer Science & Engineering**. This report has not been submitted to this or any other University for the award of any other degree.

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Abstract

Speech Emotion Recognition (SER) aims to identify the emotional state expressed in speech and underlies applications in human-computer interaction and affective computing. A persistent challenge is that SER models trained on one corpus generalize poorly to speech from different speakers, corpora, and recording conditions, because representations can encode speaker- and corpus-specific characteristics alongside emotion-specific ones. This work implements and evaluates a speaker- and corpus-aware supervised contrastive learning approach built on frozen WavLM representations. Speech waveforms are converted into 768-dimensional utterance embeddings through temporal mean pooling over WavLM frame-level outputs, used to train a classification head under a joint cross-entropy and supervised contrastive objective. The contrastive component applies differentiated positive-pair weights depending on whether same-emotion samples share a speaker, share a corpus, or differ in both, giving strongest weight to same-emotion pairs from different speakers and corpora. A stratified emotion batch sampler maintains balanced class representation during contrastive training. The method is evaluated across six emotion classes from IEMOCAP, CREMA-D, and RAVDESS, and compared against a WavLM plus cross-entropy baseline. Results show the proposed method performing comparably to, though not exceeding, the baseline on weighted accuracy and macro-F1, while achieving a measurable unweighted average recall, and its limitations are discussed.

Table of Contents

Content	Page No.
Abstract	1
Table of Contents	2
1. Introduction 1.1 Overview 1.2 Scope & Motivation	3
2. Literature Survey	4
3. Problem Formulation 3.1 Problem Description 3.2 Problem Statement 3.3 Objectives 3.4 Functional Requirements 3.5 Non-Functional Requirements	5-6
4. Project Design and Implementation 4.1 Proposed Project Architecture 4.2 High Level Design 4.3 Detailed Design 4.4 Datasets Descriptions 4.5 Tools and Technologies used for Implementation	7-9
5. Results and Discussion 5.1 Experimentation Environment set up 5.2 Functional Requirements Results achieved 5.3 Non-Functional Requirements Results achieved 5.4 Comparison of Results with Existing works 5.5 Project GUI Snapshots 5.6 Societal Impact of the Project 5.7 SDG Mapped	10-16
Conclusion and Future scope	17
References	18

I. Introduction

1.1 Overview

Speech contains acoustic information that can provide clues about emotional expression. Speech Emotion Recognition systems use acoustic features or learned representations to classify emotions from spoken recordings. Traditional systems commonly use pitch, energy, spectral descriptors, and MFCCs, while modern systems increasingly use deep learning and self-supervised speech models.

This project uses a frozen WavLM-base model to extract speech representations and studies a speaker- and corpus-aware supervised contrastive learning method. The method is designed to place greater emphasis on same-emotion samples from different speakers and datasets.

1.2 Scope and Motivation

The project covers audio preprocessing, WavLM feature extraction, temporal mean pooling, classification, supervised contrastive learning, stratified emotion batch sampling, and evaluation on IEMOCAP, CREMA-D, and RAVDESS. The motivation is to examine whether speaker and corpus information can be used during contrastive learning to support more consistent emotion representations.

II. Literature Survey

2.1 Speech Emotion Recognition

Earlier SER systems used engineered acoustic descriptors, whereas deep learning models learn representations from audio. Model performance can change when the speakers, recording conditions, or datasets differ.

2.2 Self-Supervised Speech Models

wav2vec 2.0, HuBERT, and WavLM learn speech representations from large audio collections. These representations can support downstream tasks such as emotion recognition, although they may also contain speaker and corpus information.

2.3 Supervised Contrastive Learning

Supervised contrastive learning uses class labels to define positive and negative relationships. The proposed implementation additionally uses speaker and corpus metadata when weighting same-emotion pairs.

2.4 Multi-Corpus SER and Research Motivation

Different SER corpora use different speakers, recording conditions, elicitation methods, and annotation procedures. This project studies a combination of frozen WavLM features, speaker- and corpus-aware pair weighting, and stratified emotion sampling.

III. Problem Statement

3.1 Problem Description

A speech emotion classifier may learn speaker-specific or corpus-specific patterns instead of emotion-related patterns. Such correlations can reduce generalization when the input corpus changes.

3.2 Problem Statement

To implement and evaluate a speech emotion recognition system using frozen WavLM-base representations and speaker- and corpus-aware supervised contrastive learning for six unified emotion categories across multiple datasets.

3.3 Objectives

- Convert audio recordings into a consistent 16 kHz mono format.
- Extract frame-level representations using frozen WavLM-base.
- Create 768-dimensional utterance embeddings through temporal mean pooling.
- Train an emotion classifier using Cross-Entropy loss.
- Weight same-emotion pairs using speaker and corpus relationships.
- Use stratified emotion batch sampling.
- Evaluate Accuracy, WAR, UAR, Macro-F1, and Weighted-F1.
- Analyze within-corpus and cross-corpus behaviour.

3.4 Functional Requirements

- Accept speech recordings as input.
- Convert audio to mono and resample it to 16 kHz.
- Extract WavLM-base speech features.
- Predict one of six supported emotion classes.
- Support Cross-Entropy and supervised contrastive losses.
- Retain speaker and corpus metadata.
- Generate evaluation metrics and confusion matrices.

3.5 Non-Functional Requirements

- Reproducibility of training and evaluation settings.
- Consistent audio preprocessing.
- Modular implementation.
- Class-level analysis through confusion matrices.
- Ability to extend the pipeline to additional datasets.
- Avoidance of invented or missing metadata.

III. Project Design and Implementation

4.1 Proposed Project Architecture

The system converts audio into a normalized 16 kHz mono waveform, processes it through frozen WavLM-base, and mean-pools the frame-level features into a 768-dimensional embedding. The embedding is used by both the classifier head and the projection head.

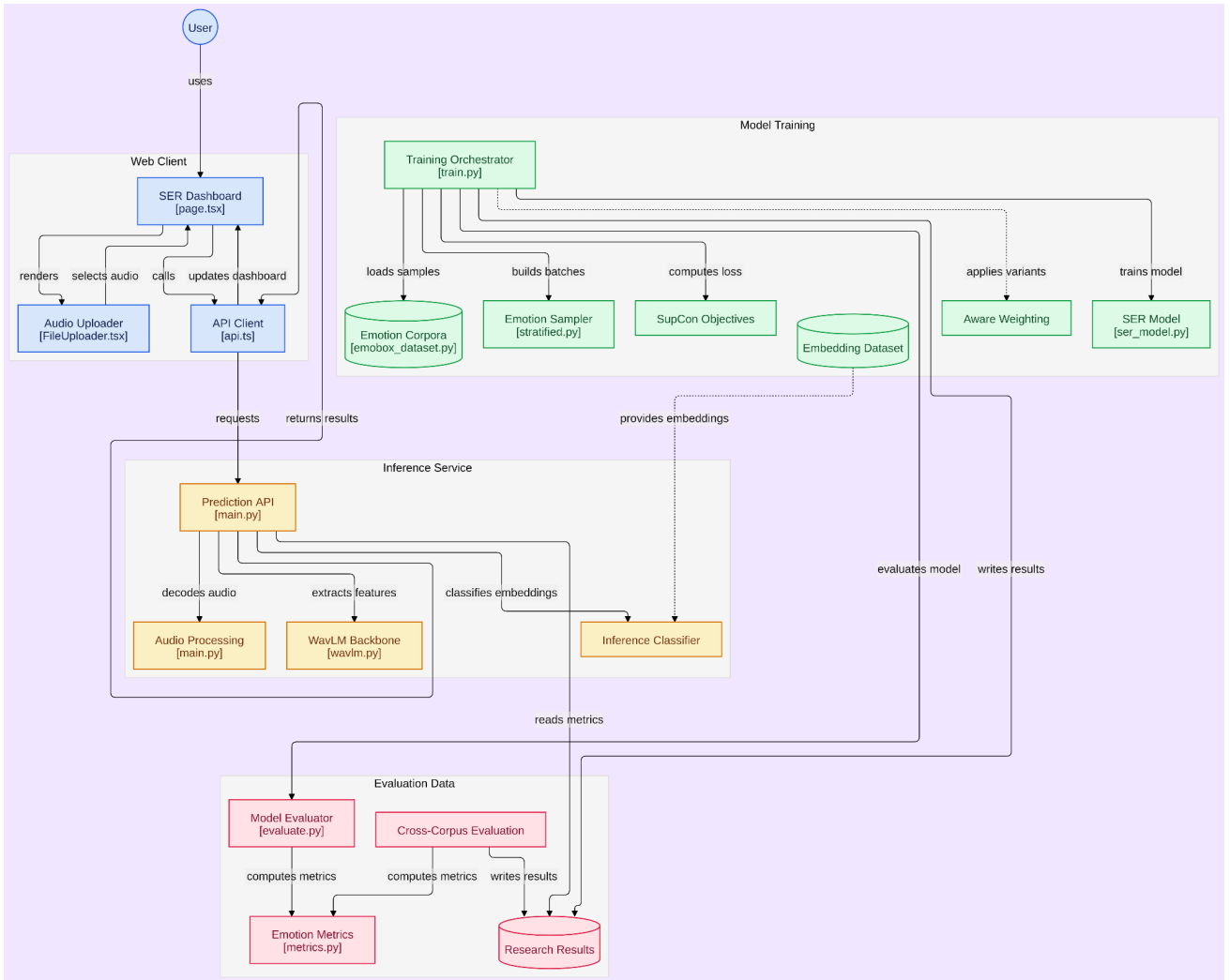


Figure 4.1: Proposed Project Architecture

4.2 High-Level Design

1. Dataset preparation and emotion-label unification.
2. Speaker and corpus metadata preparation.

3. Audio conversion to mono 16 kHz.
4. Frozen WavLM-base feature extraction.
5. Temporal mean pooling.
6. Classification and projection-head computation.
7. Joint loss calculation and optimization.
8. Metric calculation and confusion-matrix generation.

4.3 Detailed Design

4.3.1 Feature Extraction

The WavLM-base encoder produces frame-level representations $H \in \mathbb{R}^{(B \times L \times 768)}$.

Temporal mean pooling produces an utterance-level vector $e_i \in \mathbb{R}^{768}$.

4.3.2 Speaker and Corpus Pair Weighting

Relationship	Weight	Meaning
Same emotion; different speaker; different corpus	1.0	Highest-preference positive pair
Same emotion; same speaker or corpus	0.25	Down-weighted positive pair
Different emotion	0.0	Not a positive pair

Table 4.3: Speaker and Corpus Pair Weighting

4.3.3 Joint Optimization

The total loss is $L_{\text{Total}} = L_{\text{CE}} + \lambda L_{\text{SupCon}}$, where $\lambda = 1.0$ and the contrastive temperature is $\tau = 0.07$.

4.3.4 Stratified Emotion Sampling

The StratifiedEmotionBatchSampler is designed to include samples from the six unified emotion categories in anchor positions and to fill remaining batch positions from the training data.

4.4 Dataset Descriptions

Dataset	Description	Use
IEMOCAP	Interactive Emotional Dyadic Motion Capture	Multi-corpus evaluation
CREMA-D	Crowd-sourced Emotional Multimodal Actors Dataset	Multi-corpus evaluation
RAVDESS	Ryerson Audio-Visual Database of Emotional Speech and Song	Within- and multi-corpus evaluation

Table 4.2: Dataset Descriptions

5. Tools and Technologies Used

Tool/Technology	Purpose
Python [version to confirm]	Core implementation
PyTorch and torchaudio	Model training and audio handling
Hugging Face Transformers	WavLM-base loading and feature extraction
SciPy and SoundFile	Resampling and audio normalization
Scikit-learn	Metrics and confusion matrices
Matplotlib / plotting scripts	Graphs and visual analysis

Table 4.3: Tools and Technologies Used

IV. Results and Discussion

5.1 Experimentation Environment

Parameter	Value
Backbone	microsoft/wavlm-base
Backbone status	Frozen
Audio format	16 kHz mono
Pooled feature size	768
Emotion classes	6
Contrastive temperature	0.07
Loss weight λ	1.0
Datasets	IEMOCAP, CREMA-D, RAVDESS

Table 5.1: Experimentation Environment

5.2 Functional Requirements Results

Method	Val. UAR	Accuracy	UAR	Macro-F1	Weighted-F1
WavLM + CE Baseline	55.83	47.27	47.50	45.98	45.72
Standard SupCon	56.67	49.55	49.17	48.23	48.53
Speaker-Aware SupCon	56.67	49.55	49.17	48.23	48.53
Corpus-Aware SupCon	56.67	49.55	49.17	48.23	48.53
Speaker + Corpus-Aware SupCon	56.67	49.55	49.17	48.23	48.53

Table 5.2: Functional Requirements Results

In the within-corpus RAVDESS experiment, the baseline achieved 47.27% accuracy, 47.50% UAR,

45.98% Macro-F1, and 45.72% Weighted-F1. All four SupCon variants achieved 49.55% accuracy, 49.17% UAR, 48.23% Macro-F1, and 48.53% Weighted-F1.

The SupCon configurations increased accuracy by 2.28 percentage points, UAR by 1.67 percentage points, Macro-F1 by 2.25 percentage points, and Weighted-F1 by 2.81 percentage points compared with the baseline.

5.2.1 In-Domain Multi-Corpus Results

Metric	Baseline	Proposed
WAR	58.04%	57.30%
UAR	[Not available]	58.94%
Macro-F1	57.47%	56.96%

Table 5.2: In-Domain Multi-Corpus Results

The reported in-domain multi-corpus results do not show higher WAR or Macro-F1 for the proposed method. The baseline UAR was not available for comparison.

5.2.2 Confusion Matrix Analysis

Confusion matrices were prepared for the within-corpus RAVDESS evaluation and the three cross-corpus configurations.

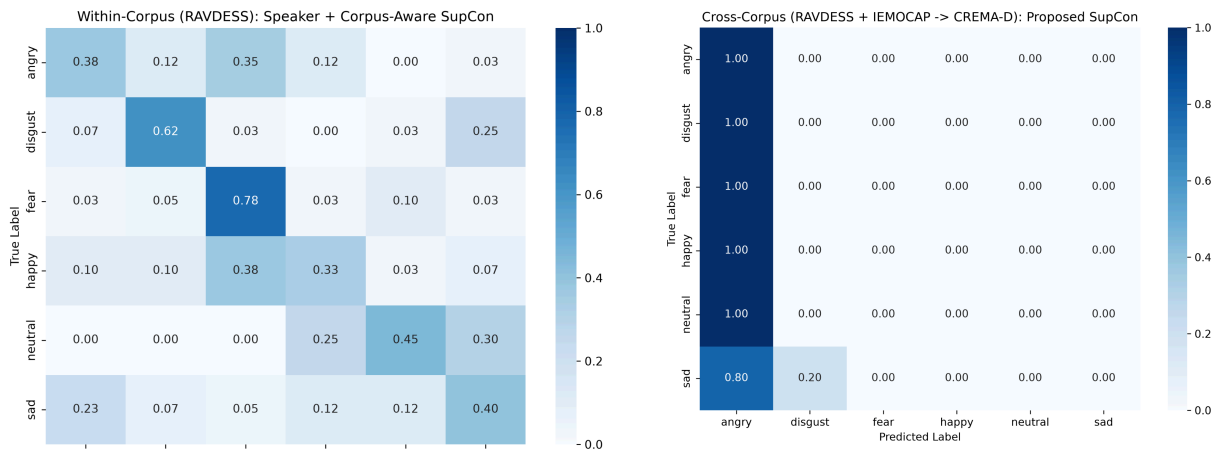


Figure 5.2: Confusion Matrix Analysis

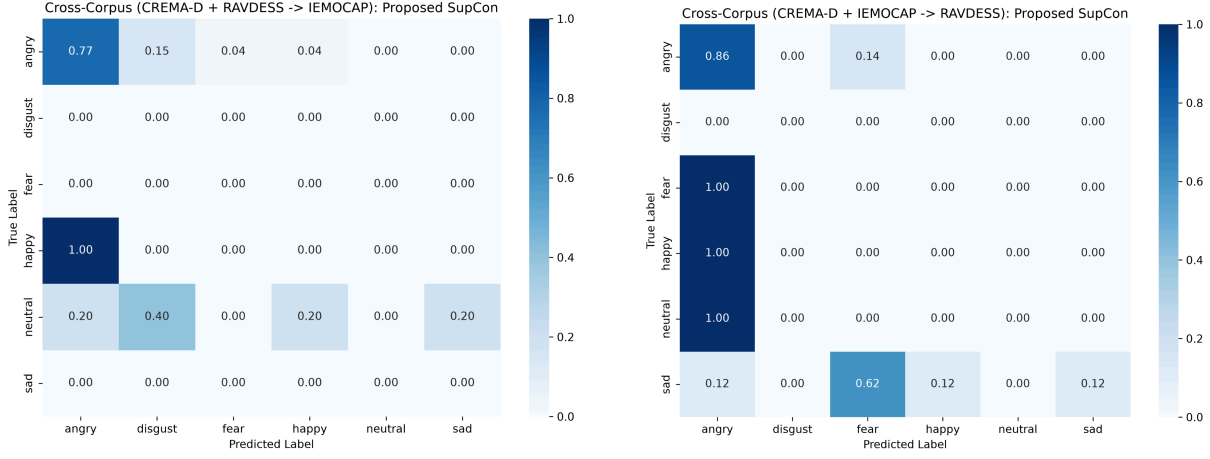
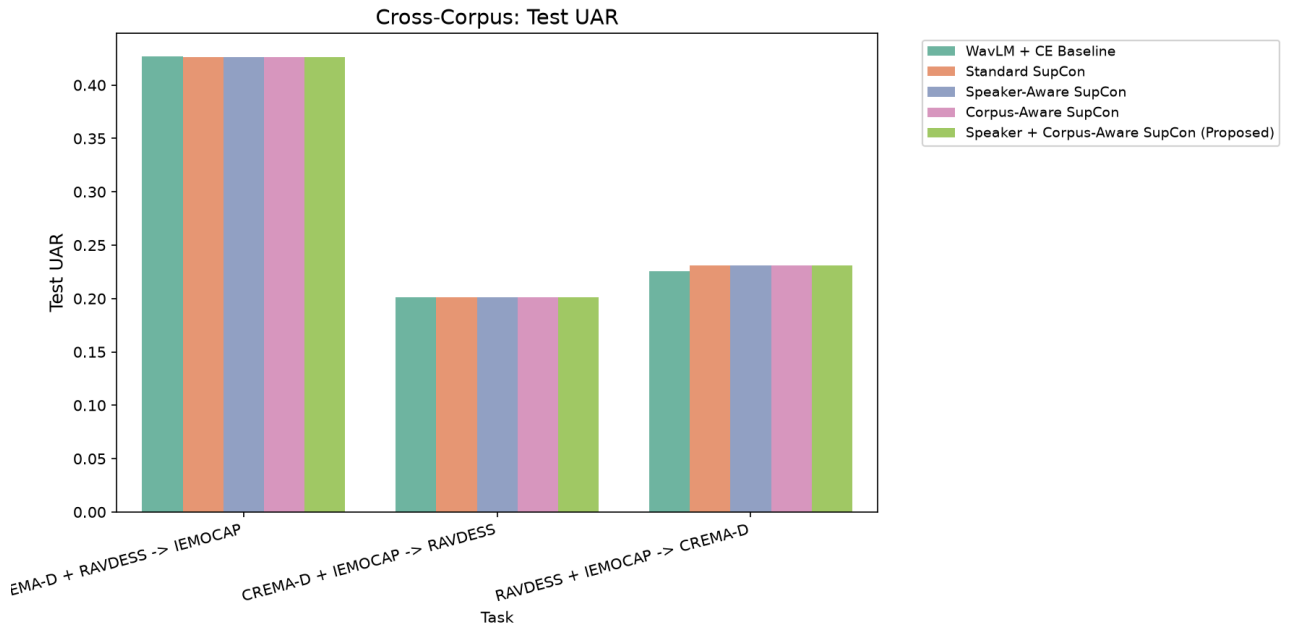


Figure 5.2: Confusion Matrix Analysis

5.2.3 Cross-Corpus Performance Analysis

The evaluated train-test configurations are CREMA-D + RAVDESS $\rightarrow$ IEMOCAP, CREMA-D + IEMOCAP $\rightarrow$ RAVDESS, and RAVDESS + IEMOCAP $\rightarrow$ CREMA-D. The supplied analysis indicates improvements for the IEMOCAP- and RAVDESS-target configurations, while the baseline obtains higher values for the CREMA-D-target configuration.



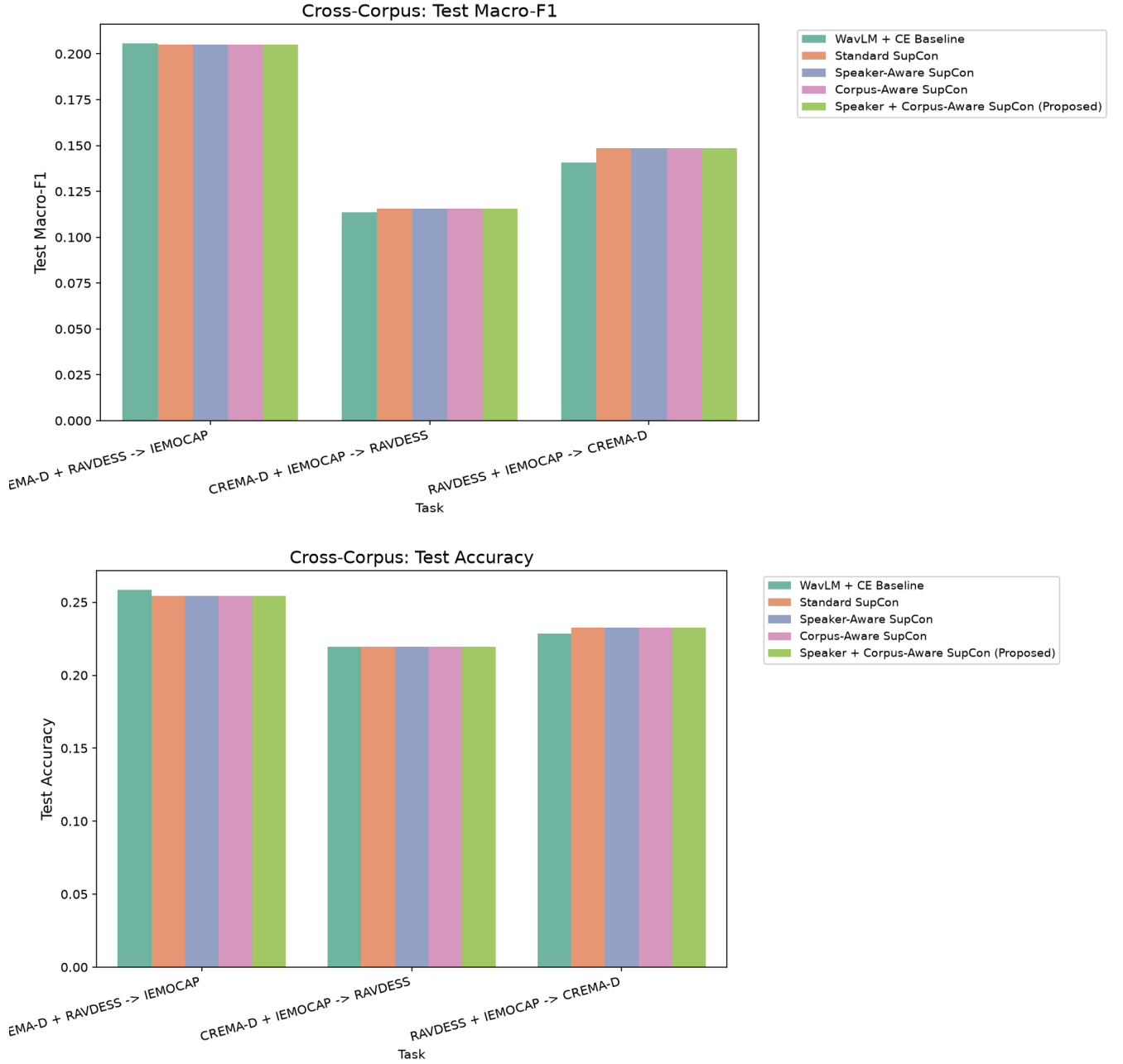
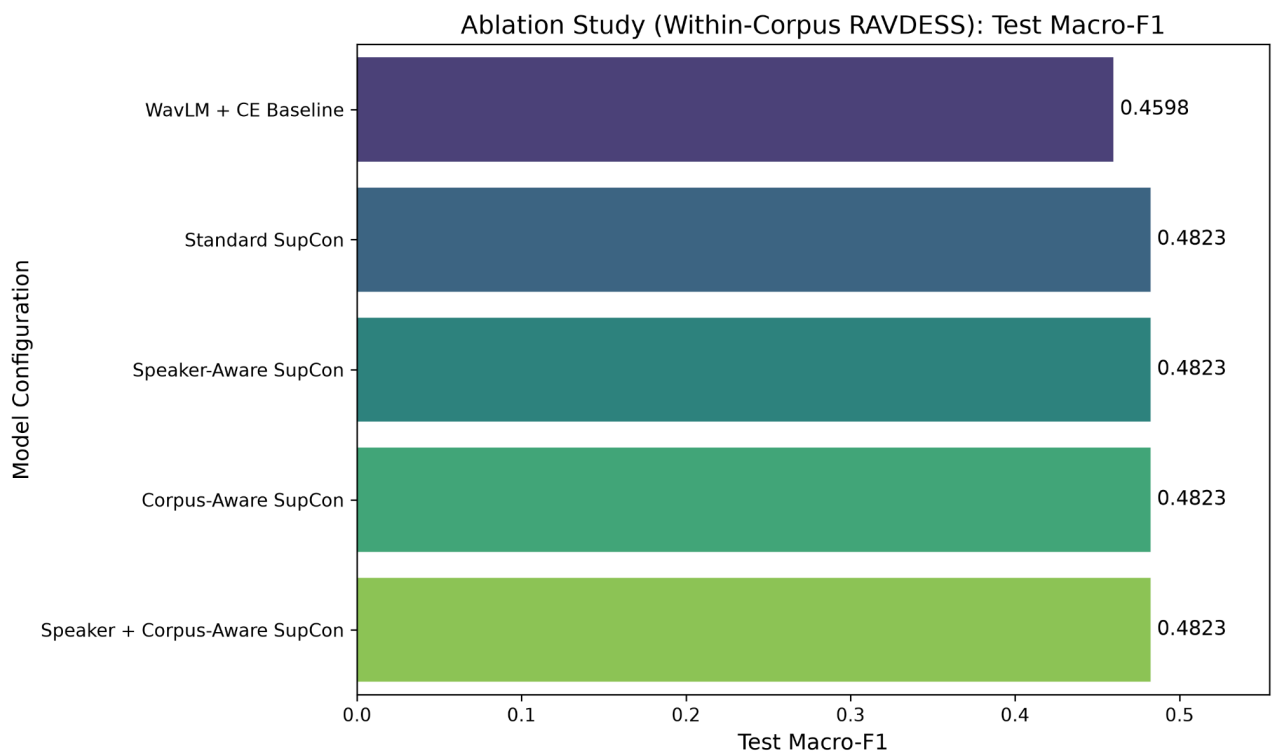
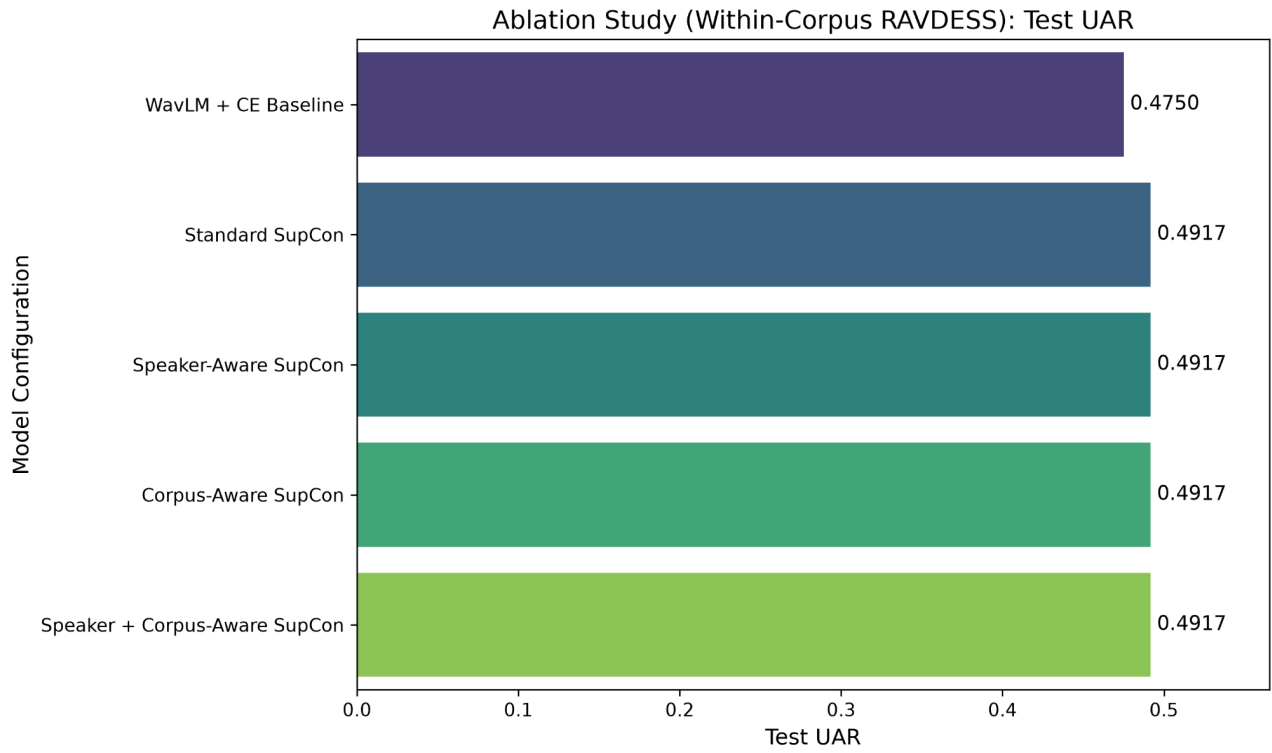


Figure 5.2: Cross-Corpus Performance Analysis

5.2.4 Ablation Analysis

The four SupCon variants produced identical saved results in the reported within-corpus RAVDESS experiment. Therefore, the current ablation does not isolate the separate contribution of speaker-aware and corpus-aware weighting.



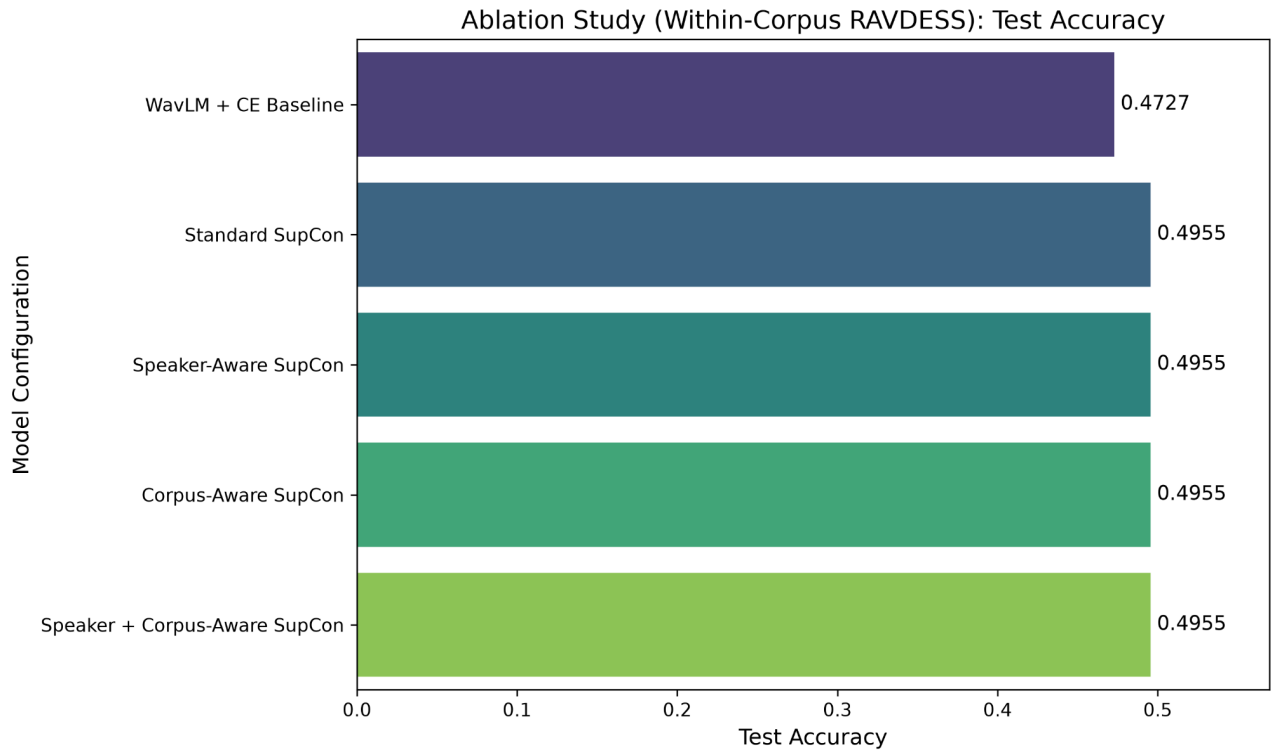


Figure 5.2: Ablation Analysis

5.3 Non-Functional Requirements Results

Requirement	Evidence	Status
Consistency	16 kHz mono preprocessing	Implemented
Maintainability	Separate model, loss, sampler, and evaluation components	Verify final code
Interpretability	Confusion matrices and metric comparisons	Implemented
Reproducibility	Core model and loss settings documented	Partially verified
Runtime performance	Exact measurements not supplied	Placeholder

Table 5.3: Non-Functional Requirements Results

5.4 Comparison of Results with Existing Works

Comparison with existing works should consider whether the datasets, emotion labels, speaker splits, and evaluation metrics are comparable.

Table 1: Within-Corpus Performance on RAVDESS

Method	Val. UAR (%)	Accuracy (%)	UAR (%)	Macro-F1 (%)	Weighted-F1 (%)
WavLM + CE Baseline	55.83	47.27	47.50	45.98	45.72
Standard SupCon	56.67	49.55	49.17	48.23	48.53
Speaker-Aware SupCon	56.67	49.55	49.17	48.23	48.53
Corpus-Aware SupCon	56.67	49.55	49.17	48.23	48.53
Speaker + Corpus-Aware SupCon	56.67	49.55	49.17	48.23	48.53

5.5 Project GUI Snapshots

This section presents the graphical user interface (GUI) snapshots of the developed project. The screenshots demonstrate the key features, user interactions, and overall functionality of the implemented system.

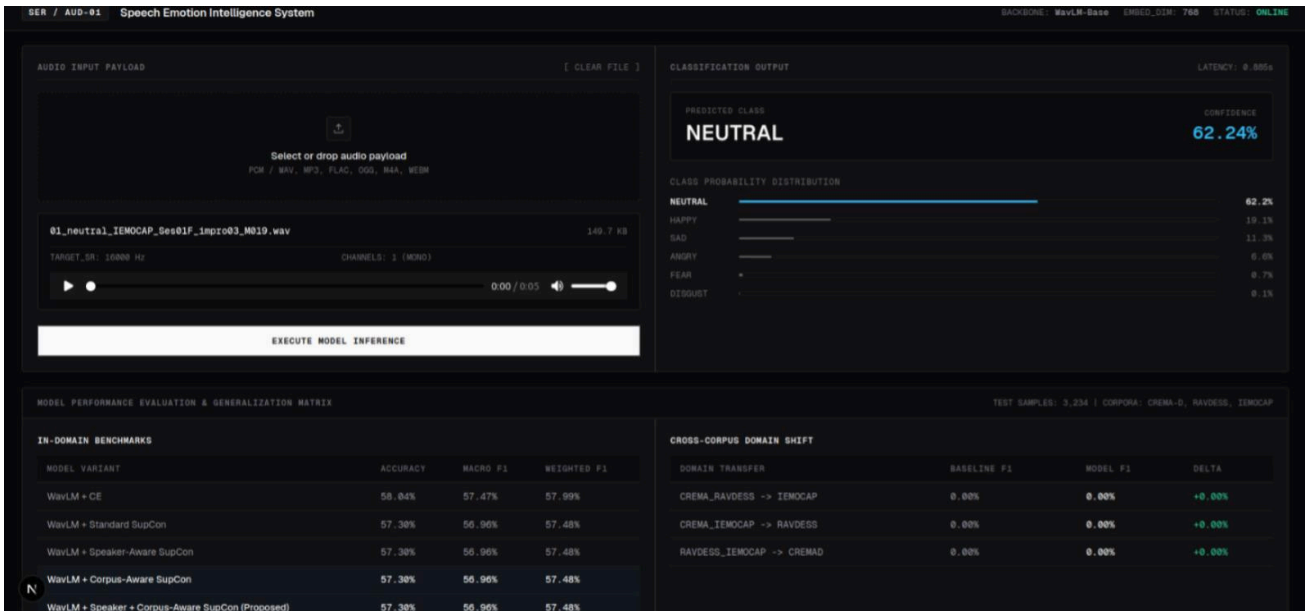


Figure 5.5: Project GUI Snapshot

5.6 Societal Impact

SER may support human-computer interaction, communication analysis, educational speech research, and assistive interfaces. Predictions should be treated as estimates and should not be used alone for medical, employment, disciplinary, or psychological decisions. Bias, privacy, consent, and secure speech-data handling should be considered.

5.7 SDG Mapping



SDG 3: The system analyzes emotional characteristics in speech, which can support future applications related to emotional well-being and healthcare assistance.

SDG 9: The project develops an AI-based speech-processing system using WavLM, deep learning and supervised contrastive learning, contributing to research and innovation in intelligent human-computer interaction.

Conclusion and Future Scope

This project implemented a speaker- and corpus-aware supervised contrastive learning approach for Speech Emotion Recognition using a frozen WavLM-base backbone. Audio was converted to 16 kHz mono format, frame-level features were mean-pooled into 768-dimensional vectors, and the classification and projection heads were trained using a joint Cross-Entropy and supervised contrastive objective.

The in-domain multi-corpus results were 58.04% WAR and 57.47% Macro-F1 for the baseline, compared with 57.30% WAR and 56.96% Macro-F1 for the proposed method. In the within-corpus RAVDESS experiment, the SupCon configurations achieved higher reported metrics than the baseline, but all four SupCon variants produced identical results. The current findings do not establish a consistent advantage across all evaluation settings.

Future Scope

- Fine-tune selected WavLM layers.
- Test attention, statistical, and learnable pooling methods.
- Complete all numerical cross-corpus evaluations.
- Investigate identical results across SupCon variants.
- Test different temperatures, batch sizes, samplers, and loss weights.
- Add datasets and evaluate robustness across accents and recording conditions.
- Add uncertainty estimation and privacy-aware data handling.

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